Tutorial for using `Git` and `GitHub`

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1 Foreword

High-Performance Computing (HPC) refers to the use of powerful computers that combine many processors to solve problems much faster than a typical personal computer. Instead of relying on a single CPU, HPC systems distribute work across many CPUs or GPUs, allowing complex computations to be performed efficiently.

HPC is widely used in science, engineering, medicine, and data-intensive research. For example, engineers use HPC to run large simulations, process large datasets, train machine learning models, and perform advanced optimisation or uncertainty analysis.

In universities, HPC systems are shared resources that researchers and students access remotely. Because these systems are typically operated through the command line, understanding basic Linux commands and terminal usage is an essential first step to working effectively on an HPC cluster.

2 What is HPC?

High-Performance Computing (HPC) uses many interconnected processors to solve computational problems faster than a standard computer. These systems are often organised as **clusters**, where multiple machines work together to perform tasks in parallel.

By distributing workloads across many processors, HPC systems can handle large simulations, data analysis tasks, and computational models that would otherwise take too long to run on a personal computer.

3 Why do we use HPC?

Many scientific and engineering problems require more computational power than a typical laptop or desktop can provide. HPC enables researchers to perform large-scale calculations efficiently.

Common reasons for using HPC include:

- Running **large simulations** such as fluid dynamics or structural modelling
- **Reducing computation time** through parallel processing
- Handling **large datasets** in scientific or data-driven research
- Training **machine learning and AI models** that require significant computing resources

4 Typical Components of an HPC System

An HPC cluster typically includes several key components:

Login Nodes

Login nodes are the entry point to the cluster. Users connect to these nodes remotely to manage files, prepare scripts, and submit jobs.

Compute Nodes

Compute nodes perform the actual computations. Jobs submitted by users are scheduled to run on these machines.

Storage Systems

Shared storage allows users to access data and results from any node in the cluster.

Job Scheduler

A scheduler distributes jobs across available resources such as CPUs, memory, and GPUs. Common schedulers include **SLURM**, **PBS**, and **LSF**.

5 How Users Access an HPC Cluster

Users typically access an HPC cluster remotely using **Secure Shell (SSH)**. After logging in, they interact with the system through a terminal to manage files, load software modules, and submit jobs to the scheduler.

To connect to the HPC cluster, open a terminal application. This can be:

- **Terminal** on macOS or Linux
- **Command Prompt** or **PowerShell** on Windows

Then run the following command:

```
ssh <id>@cognition.cose.gla.ac.uk
```

where `<id>` should be replaced with your university or HPC username.

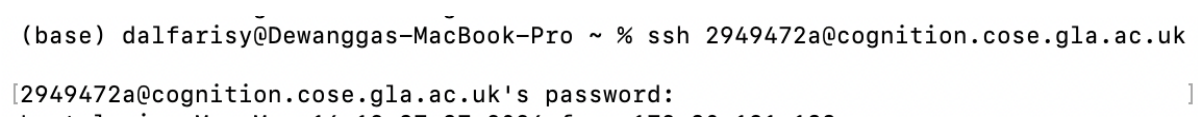

Figure 1: Example SSH login to the HPC cluster

After running the command, you will be prompted to enter your password. While typing the password, **no characters or cursor movement will appear on the screen**. This is normal behaviour in Linux systems for security reasons. Simply type your password carefully and press **Enter**.

During the first login, you may be asked to reset your password. If prompted:

1. Enter your **current password** (the one used to log in).
2. Enter a **new password**.
3. Re-enter the new password to confirm.

Once the password is successfully updated, you will be logged out of the HPC system. Simply put again the ssh sign and re-login with the new password.

6 Basic Linux and Terminal Skills for HPC

Most HPC systems run on Linux and are operated through the command line (Command Prompt for Windows, Terminal for Mac and Linux). Therefore, basic familiarity with Linux commands is essential.

Key skills include:

6.1 File System Navigation

When working on an HPC system, most interaction happens through the **file system**.
Understanding how to check your location and move between directories is essential.

6.1.1 Check Your Current Location (`pwd`)

The command `pwd` stands for **print working directory**. It tells you where you currently are in the file system.

Example:

```
pwd
```

Example output:

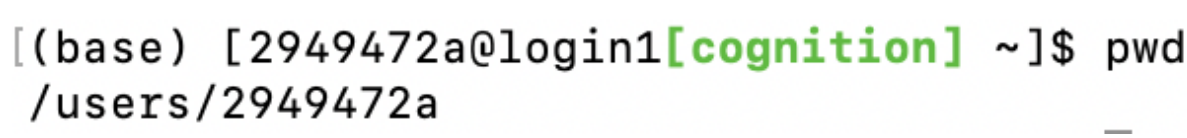

Figure 2: Location checking using pwd

This means you are currently inside your **home directory**.

It is good practice to run `pwd` when you are unsure where you are in the file system.

6.1.2 List Files and Folders (`ls`)

The `ls` command shows the contents of the current directory.

Example:

```
ls
```

Example output:

Figure 3: List files and folders

Other significant command is to add -l for showing additional informations.

Example:

```
ls -l
```

Example output:

Figure 4: List of Files and Folders with Additional Informations

This shows additional information such as:

- file permissions
- file size
- modification time

6.1.3 Change Directory (`cd`)

The `cd` command allows you to move between directories.

Example:

```
cd Test
```

Example output:

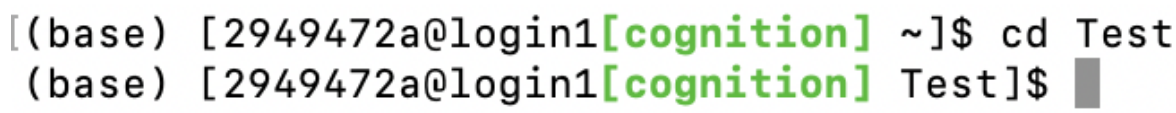

Figure 5: Moving directories

6.2 File Management

File management allows users to create directories, copy files, move files, and remove files within the HPC system. These commands are essential for organising data and managing project files.

The most commonly used commands are:

- `mkdir` – create a new directory
- `cp` – copy files or directories
- `mv` – move or rename files
- `rm` – remove files or directories
- `sftp` – transfer files between your computer and the HPC system

6.2.1 Create a Directory (`mkdir`)

The `mkdir` command creates a new folder. The `nano` command create a new text file for our tests.

```
mkdir newFolder
nano text1.txt
```

When using the nano command, you will encounter movement to different UI, dont worry and for now, just exit the UI.

You can confirm the new folder was created by

```
ls
```

Example output:

```
[(base) [2949472a@login1[cognition] ~]$ ls
[archive  localscratch  newFolder  sharedscratch  Test
[(base) [2949472a@login1[cognition] ~]$ nano text1.txt
[(base) [2949472a@login1[cognition] ~]$ ls
 archive  localscratch  newFolder  sharedscratch  Test  text1.txt
```

Figure 6: Making new directory

6.2.2 Copy Files (`cp`)

The `cp` command copies a file from one location to another.

Example:

```
cp text1.txt text2.txt
```

Example output:

```
[(base) [2949472a@login1[cognition] ~]$ cp text1.txt text2.txt                    ]
[(base) [2949472a@login1[cognition] ~]$ ls                                        ]
 archive  localscratch  newFolder  sharedscratch  Test  text1.txt  text2.txt
```

Figure 7: Copy files to different file name

This creates a copy of `text1.txt` called `text2.txt`.

You can also copy a file into a directory.

Example:

```
cp text1.txt newFolder/
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ cp text1.txt newFolder/
(base) [2949472a@login1[cognition] ~]$ ls
 archive  localscratch  newFolder  sharedscratch  Test  text1.txt  text2.txt
(base) [2949472a@login1[cognition] ~]$ cd newFolder
(base) [2949472a@login1[cognition] newFolder]$ ls
 text1.txt
```

Figure 8: Copy files to folder

This creates a copy of `text1.txt` inside `newFolder`.

6.2.3 Move or Rename Files (`mv`)

The `mv` command moves files between directories or renames them.

Rename a file:

```
mv text1.txt newfile.txt
```

Example output:

```
(base) [2949472a@login1[cognition] newFolder]$ mv text1.txt newfile.txt
(base) [2949472a@login1[cognition] newFolder]$ ls
 newfile.txt
```

Figure 9: Rename files

This renames `text1.txt` to `newfile.txt`

Move a file to directory:

```
mv text2.txt newFolder/
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ mv text2.txt newFolder/
(base) [2949472a@login1[cognition] ~]$ cd newFolder
(base) [2949472a@login1[cognition] newFolder]$ ls
 newfile.txt  text2.txt
```

Figure 10: Moving files to folder

This moves `text2.txt` to `newFolder`

6.2.4 Remove Files (`rm`)

The `rm` command removes files.

Example:

```
rm text1.txt
```

Example output:

```
(base) [2949472a@login1[cognition] newFolder]$ rm text2.txt
(base) [2949472a@login1[cognition] newFolder]$ ls
newfile.txt
```

Figure 11: Remove files

This removes `text2.txt`.

To remove a folder, add -r to the given command.

Example:

```
mkdir removable-folder
ls
rm -r removable-folder
```

Example output:

```
(base) [2949472a@login1[cognition] newFolder]$ mkdir removable-folder
(base) [2949472a@login1[cognition] newFolder]$ ls
newfile.txt  removable-folder
(base) [2949472a@login1[cognition] newFolder]$ rm -r removable-folder
(base) [2949472a@login1[cognition] newFolder]$ ls
newfile.txt
```

Figure 12: Remove folder

This removes `removable-folder` and all corresponding files inside.

Important:
Files deleted with `rm` are permanently removed.
There is **no recycle bin or undo option** on most HPC systems.

Always confirm your location before deleting files:

```
pwd
```

6.2.5 Transfer Files Using `sftp`

Files can be transferred between your local computer and the HPC cluster using `sftp`.

Start an SFTP session:

```
sftp <id>@cognition.cose.gla.ac.uk
```

Once connected, you can move your files:

```
put <file directory>
```

As an example, we want to put the helloworld.ipnyb in our training-files to the hpc system from our local computer.

Example code:

```
put <file directory>/hello-world.ipynb
```

Example output:

```
sftp> put /Users/dalfarisy/Library/CloudStorage/OneDrive-UniversityofGlasgow/PhD
-related/quantum-ml-pqc/uog-hpc/training-files/hello-world.ipynb
Uploading /Users/dalfarisy/Library/CloudStorage/OneDrive-UniversityofGlasgow/PhD
-related/quantum-ml-pqc/uog-hpc/training-files/hello-world.ipynb to /users/29494
72a/hello-world.ipynb
hello-world.ipynb                             100%  255    42.0KB/s   00:00
```

Figure 13: Practice putting files using SFTP.

Or download your file from HPC:

As an example, we want to download the same file into our local computer.

```
get <file/folder in hpc> <file/folder directory>
```

Example output:

```
sftp> get hello-world.ipynb /Users/dalfarisy/Documents
Fetching /users/2949472a/hello-world.ipynb to /Users/dalfarisy/Documents/hello-w
orld.ipynb
hello-world.ipynb                             100%  255    22.9KB/s   00:00
```

Figure 14: Practice downloading files using SFTP.

6.2.6 Practice Example

Here are simple practice example for familiarise yourself with Linux system.

Example code:

```
mkdir testfolder
nano text1.txt
ls
cp text1.txt testfolder/
mv text1.txt file_backup.txt
ls
rm file_backup.txt
ls
cd testfolder
ls
```

Example output:

```
(base) [2949472a@login1[cognition] newFolder]$ mkdir testfolder
(base) [2949472a@login1[cognition] newFolder]$ nano text1.txt
(base) [2949472a@login1[cognition] newFolder]$ ls
testfolder  text1.txt
(base) [2949472a@login1[cognition] newFolder]$ cp text1.txt testfolder/
(base) [2949472a@login1[cognition] newFolder]$ mv text1.txt file_backup.txt
(base) [2949472a@login1[cognition] newFolder]$ ls
file_backup.txt  testfolder
(base) [2949472a@login1[cognition] newFolder]$ rm file_backup.txt
(base) [2949472a@login1[cognition] newFolder]$ ls
testfolder
(base) [2949472a@login1[cognition] newFolder]$ cd testfolder
(base) [2949472a@login1[cognition] testfolder]$ ls
text1.txt
(base) [2949472a@login1[cognition] testfolder]$
```

Figure 15: Practice example code and its given output

These commands will help you practise creating directories, copying files, moving files, and removing files.

In order to use this training, we upload hpc-text.txt that is shared in this github using SFTP by using:

```
sftp <id>@cognition.cose.gla.ac.uk
put <directory>/hpc-text.txt
```

6.3 Adding texts to file

6.3.1 Using echo to create files

We can use echo to create text files,

Example:

```
echo "This is a test"
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ echo "This is a test."
This is a test.
```

Figure 16: Echo some words

Yes, echo just prints its arguments back out again (hence the name). But combine it with a redirect, and you've got a way to easily create small test files:

Now we are saving the words we put in echo as a txt file.

Example:

```
echo "This is a test" > text1.txt
echo "This is a second test" > text2.txt
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ echo "This is a test." > test1.txt
(base) [2949472a@login1[cognition] ~]$ echo "This is a second test." > test2.txt
```

Figure 17: Echo saving the words as txt

We also can link both files together as one, this is called concatenate. The first two combined will concatenate the file together and the last > will act as saving to a new file.

Example:

```
cat text_1.txt text_2.txt > cat.txt
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ cat test1.txt test2.txt > cat.txt
```

Figure 18: Concatenate using cat to combine both files as new txt

6.3.2 Using nano to modify file

Using nano is much more straightforward as we can directly edit the file in the nano ecosystem and we can do various things on this environment too. We can directly edit, modify and save the files using this software.

```
nano test_1.txt
```

Example output:

```
  GNU nano 5.6.1                                test1.txt
This is a test.

                                  [ Read 1 line ]
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^_ Go To Line M-E Redo     M-6 Copy
```

Figure 19: Nano user interface in terminal.

6.4 Viewing File Contents

Sometimes it is useful to inspect the contents of a file directly from the terminal without opening a full text editor. Linux provides several commands that allow users to quickly view files.

Common commands include:

- `cat` – display the entire file
- `less` – view a file page by page
- `head` – show the first lines of a file
- `tail` – show the last lines of a file

6.4.1 Display an Entire File (`cat`)

The `cat` command prints the full contents of a file directly in the terminal.

```
cat hpc-text.txt
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ cat hpc-text.txt
The Rise of High-Performance Computing

 High-performance computing (HPC) refers to the use of powerful computer systems designed to process extremely large volumes of data and per
form complex calculations at very high speeds. Unlike conventional desktop computers, HPC systems typically combine hundreds or thousands o
f processors that work together to solve scientific, engineering, and data-intensive problems. This capability has transformed modern resea
rch by enabling simulations and analyses that were previously impossible.

One of the most significant advantages of HPC is its ability to perform parallel computation. In traditional computing, tasks are often exe
cuted sequentially, meaning each calculation must wait for the previous one to finish. In contrast, HPC systems divide large problems into
 smaller tasks that can be processed simultaneously across many computing units. This parallel approach dramatically reduces the time requir
ed to complete large calculations.

 HPC has become an essential tool in a wide range of disciplines. In climate science, researchers use HPC to simulate atmospheric and ocean
systems in order to understand long-term climate patterns. These simulations involve enormous datasets and require sophisticated mathematic
al models. Without HPC systems, running such simulations could take years rather than days or weeks.

Similarly, engineers rely on HPC to perform advanced simulations of physical systems. For example, aerospace engineers use computational mo
dels to study airflow around aircraft structures, while civil engineers simulate stress distribution in bridges and buildings. By analysing
 these models on HPC clusters, researchers can test designs under many different conditions before constructing physical prototypes.

Another rapidly growing application of HPC is in artificial intelligence and machine learning. Training large machine learning models often
 requires processing millions or even billions of data points. Graphics processing units (GPUs) and other accelerators in HPC clusters allo
w these models to be trained far more efficiently than on standard computers.

Despite these benefits, working with HPC systems requires specialised knowledge. Users must typically interact with the system through a co
mmand-line interface and submit jobs to a scheduler that manages the allocation of computational resources. This workflow is different from
 the familiar graphical interfaces used on personal computers, and new users often need training to become comfortable with it.

Most HPC clusters are shared environments used by many researchers simultaneously. To ensure fair access, resource management software sche
dules computational jobs and distributes workloads across available machines. Systems such as SLURM, PBS, and LSF are commonly used for thi
s purpose. By controlling how resources are allocated, these schedulers prevent individual users from monopolising the system.

Looking ahead, the importance of HPC is expected to grow even further. As scientific questions become more complex and datasets continue to
 expand, researchers will increasingly depend on powerful computing infrastructure to support discovery and innovation. In fields ranging f
rom medicine to astrophysics, HPC is likely to remain a central component of m(base) [2949472a@log(base) [2949472a@login1[cognition] ~(base
```

Figure 20: Viewing entire file using cat

This command works well for **small files**, but it may flood the screen if the file is very large.

6.4.2 View Files Page by Page (`less`)

The `less` command allows you to scroll through a file interactively.

Inside `less`, you can:

- press **Space** to scroll down
- press **b** to scroll up
- press **q** to quit

This is the preferred command for viewing **large files**.

Example:

```
less hpc-text.txt
```

Example output:

```
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Looking ahead, the importance of HPC is expected to grow even further. As scientific questions become more complex and datasets continue to
 expand, researchers will increasingly depend on powerful computing infrastructure to support discovery and innovation. In fields ranging f
rom medicine to astrophysics, HPC is likely to remain a central component of modern scientific progress.
hpc-text.txt (END)
```

Figure 21: Viewing entire file using less

6.4.3 View the Beginning of a File (`head`)

The `head` command shows the **first 10 lines** of a file.

Example:

```
head hpc-text.txt
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ head hpc-text.txt
The Rise of High-Performance Computing

High-performance computing (HPC) refers to the use of powerful computer systems designed to process extremely large volumes of data and per
form complex calculations at very high speeds. Unlike conventional desktop computers, HPC systems typically combine hundreds or thousands o
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dels to study airflow around aircraft structures, while civil engineers simulate stress distribution in bridges and buildings. By analysing
 these models on HPC clusters, researchers can test designs under many different conditions before constructing physical prototypes.
```

Figure 22: Viewing beginning of a file using head

You can specify the number of lines:

```
head -n 5 file.txt
```

6.4.4 View the End of a File (`tail`)

The `tail` command shows the **last 10 lines** of a file.

Example:

```
tail hpc-text.txt
```

Example output:

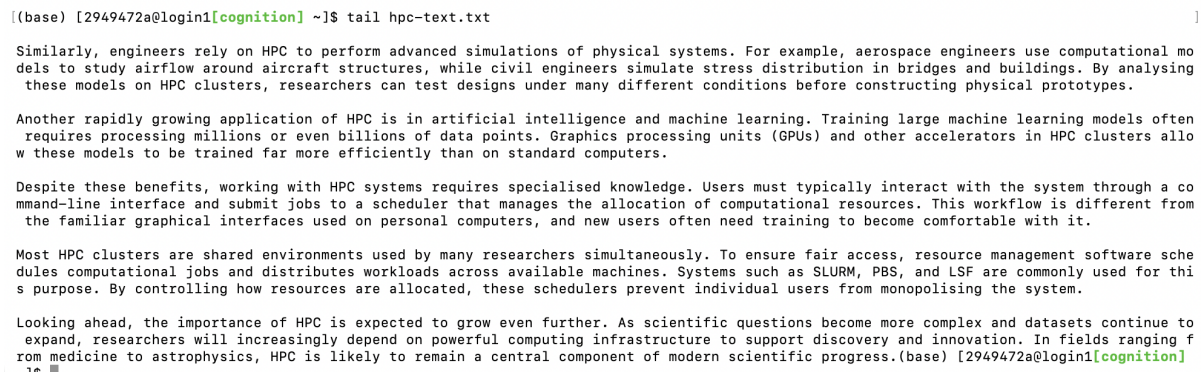


Figure 23: Viewing end of file using tail

You can specify the number of lines:

```
tail -n 5 file.txt
```

These commands allow you to quickly inspect files without opening a text editor.

One key example is to take the last data in the log file, which is coming quite handy in some analysis:

```
tail -f output.log
```

6.4.5 Practice Exercise

Try the following commands:

```
cat hpc-text.txt
less hpc-text.txt
head hpc-text.txt
tail hpc-text.txt
```

These commands allow you to quickly inspect files without opening a text editor and feel the difference between each methods.

6.5 Running Programs

After navigating files and managing directories, the next step in using an HPC system is running programs. Programs can be executed directly from the terminal or submitted as jobs to the scheduler.

In general:

- small tests or simple scripts can be run directly in the terminal
- larger computations should be submitted to the scheduler

6.5.1 Running a Script Directly using python

First, we create a simple script using nano. The file should directly printing "Hello world!" as a simple example.

Example:

```
nano testrun.py
```

Example output:

Figure 24: Making new py files.

In nano user interface, we use simple command to print "Hello world!".

Example:

```
print("Hello world!")
```

Example output:

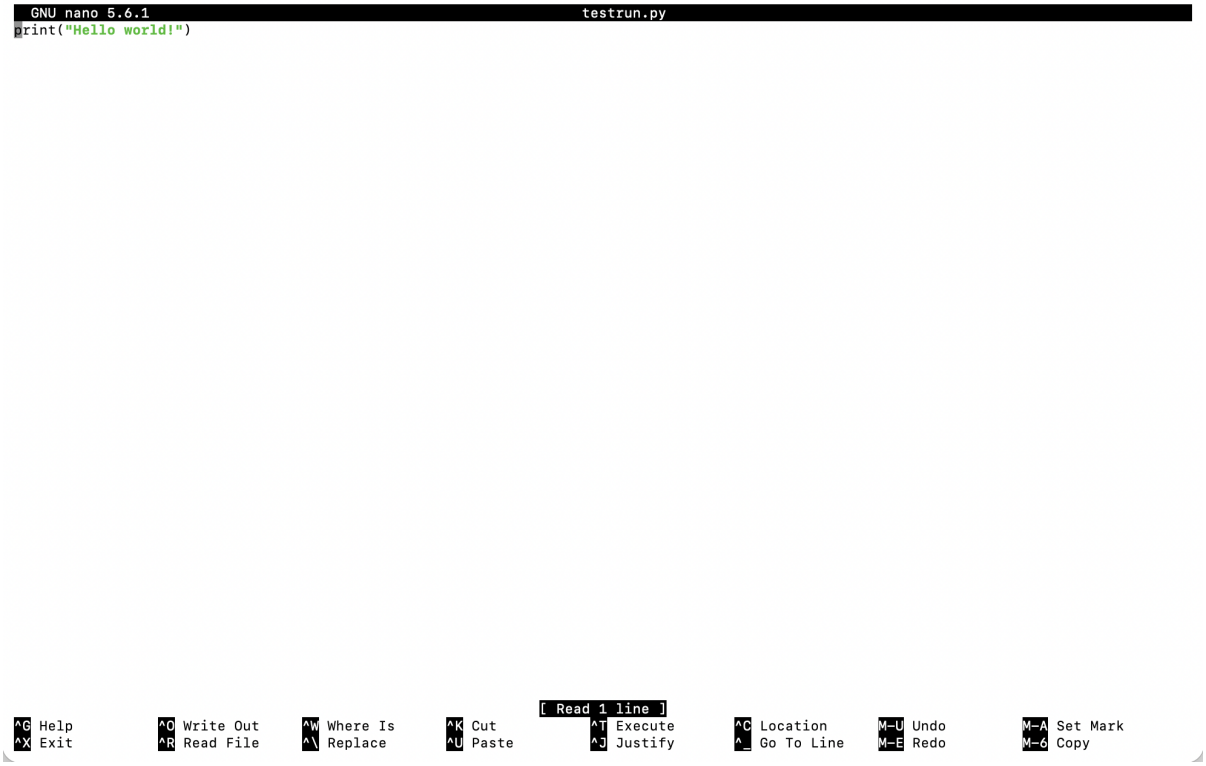


Figure 25: Writing up hello world in nano.

Then, we can check the hpc system if they have a python installed and we can use their python. We can check the module by using the following code

Example:

```
module avail
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ module avail

------------------------------------------------------- /opt/apps/eb/modules/compiler -------------------------------------------------------
   GCC/15.2.0    GCCcore/15.2.0

------------------------------------------------------- /opt/apps/eb/modules/devel ----------------------------------------------------------
   M4/1.4.19    M4/1.4.20-GCCcore-15.2.0    M4/1.4.20 (D)

------------------------------------------------------- /opt/apps/eb/modules/lang -----------------------------------------------------------
   Bison/3.8.2-GCCcore-15.2.0    Bison/3.8.2 (D)    flex/2.6.4-GCCcore-15.2.0    flex/2.6.4 (D)

------------------------------------------------------- /opt/apps/eb/modules/lib ------------------------------------------------------------
   zlib/1.3.1-GCCcore-15.2.0    zlib/1.3.1 (D)

------------------------------------------------------- /opt/apps/eb/modules/tools ----------------------------------------------------------
   EasyBuild/5.1.2    binutils/2.45-GCCcore-15.2.0    binutils/2.45 (D)    help2man/1.49.3-GCCcore-15.2.0

----------------------------------------- /opt/gridware/subscribed/etc/modulefiles/proprietary -----------------------------------------
   ansys/2025.2    comsol/6.3    matlab/r2025a

------------------------------------- /opt/gridware/subscribed/etc/modulefiles/linux-rocky9-x86_64_v4 -------------------------------------
   apptainer/1.4.4/gcc-15.2.0/a-a2wyui6        gcc/15.2.0/gcc-11.5.0/a-mffqiek
   cmake/3.31.9/gcc-11.5.0/a-xc6da5e           lmod/8.7.55/gcc-11.5.0/a-sntlpdr
   cuda/12.9.1/none-none/a-j4amwzt             miniforge3/25.3.0-3/none-none/a-4lhn6xy
   cuda/13.0.2/none-none/a-eai3dlo      (D)    openfoam/2412/gcc-15.2.0/openmpi-5.0.8/a-xb2i6l4
   easybuild/4.7.0/none-none/a-j3rz2ko         r/4.5.2/gcc-15.2.0/a-lxuwftt
```

Figure 26: Loading up available module

In the cognition, we have miniforge so we can load the miniforge. Copy the file names of the miniforge like the one below (updated 17/3/2026)

Example:

```
module load miniforge3/25.3.0-3/none-none/a-4lhn6xy
```

Example output:

```
[2949472a@login1[cognition] ~]$ module load miniforge3/25.3.0-3/none-none/a-4lhn6xy
[2949472a@login1[cognition] ~]$
```

Figure 27: Loading up miniforge module

From here on, we can make conda environment as usual, for simplicity, we will use the base.

Example:

```
conda activate base
```

Example output:

```
[2949472a@login1[cognition] ~]$ module load miniforge3/25.3.0-3/none-none/a-4lhn6xy
[2949472a@login1[cognition] ~]$
```

Figure 28: Loading up conda environment.

From here, we can run the python3 file as usual.

Example:

```
python testrun.py
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ python testrun.py
Hello world!
```

Figure 29: Running python testrun.py file.

However, when trying to run the python notebook (ipynb files), we need to convert it first to a standard python notebook.

Example:

```
jupyter nbconvert --to python hello-world.ipynb
python hello-world.py
```

Example output:

```
(base) [2949472a@login1[cognition] ~]$ jupyter nbconvert --to python hello-world.ipynb
[NbConvertApp] Converting notebook hello-world.ipynb to python
[NbConvertApp] Writing 87 bytes to hello-world.py
(base) [2949472a@login1[cognition] ~]$ python hello-world.py
Hello world!
```

Figure 30: Converting and running jupyter notebook (ipynb) file.

7 Foreword

Git and GitHub has a long history and it has been developed with in-depth and sophisticated technical features under the hood. This tutorial aims to provide readers with a direct guidance for using GitHub. The material gives step-by-step instructions by which one can apply it for their work and can have an adequate starting point for self-learning of GitHub.

Like many other software-based procedures, there are multiple ways to perform a task using GitHub. In this tutorial, we steer our focus on methods which are considered as "good practice" in terms of sustainable programming development which were suggested and recommended by the Git's user community.

8 Installation of `Git`

Git goes with WSL by default, so if you are using WSL or Linux operating system, you do not have to do the installation. If you are working with Windows or Mac, you need to install Git. The instruction for installation from Git's official website is straightforward, so for this point readers are directed to the website, here, to install Git.

9 Connect your local machine to a GitHub's repository

9.1 Introduction to `terminal`

Using Git, you will mostly do the operations via a `terminal`. So, it is worth to give a brief description of the term `terminal`. For ones who are already familiar with `terminal` can skip this part.

To have a sense of what `terminal` is, it could be described in a rudimentary way as that "*A terminal is a program that allows you to interact with your computer by typing text commands instead of clicking buttons.*"

In everyday computer use, people usually interact with a graphical interface (GUI):

- clicking folders
- dragging files
- pressing buttons

A terminal provides a text-based interface to the operating system. Instead of clicking, you type commands that tell the computer what to do. For example, to list files and folders currently in the directory, in a `terminal` (assume the `bash terminal` is used), we type:

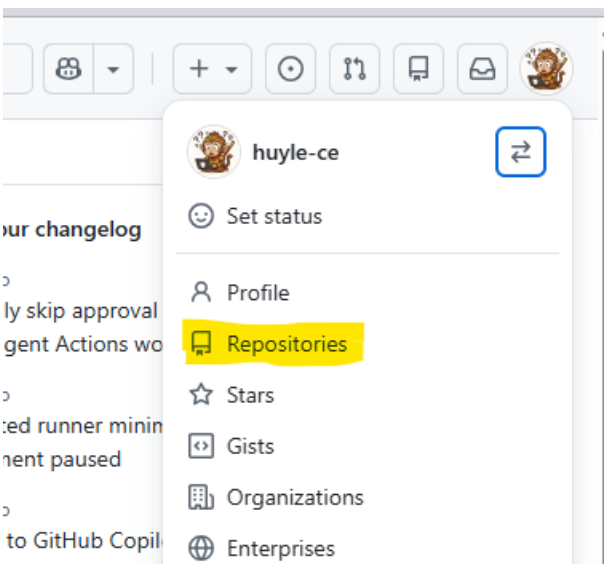
```
ls
```

and then we will see somethings like:

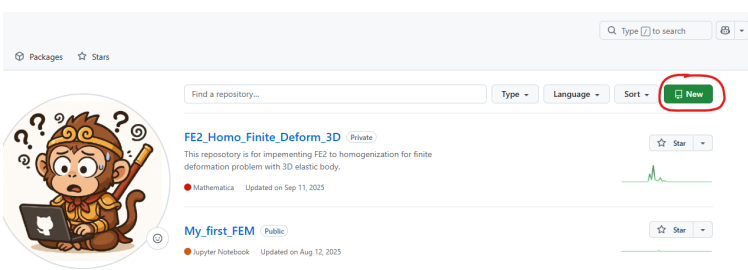
```
project
python_file.py
cpp_file.cpp
```

9.2 Create a GitHub's repository

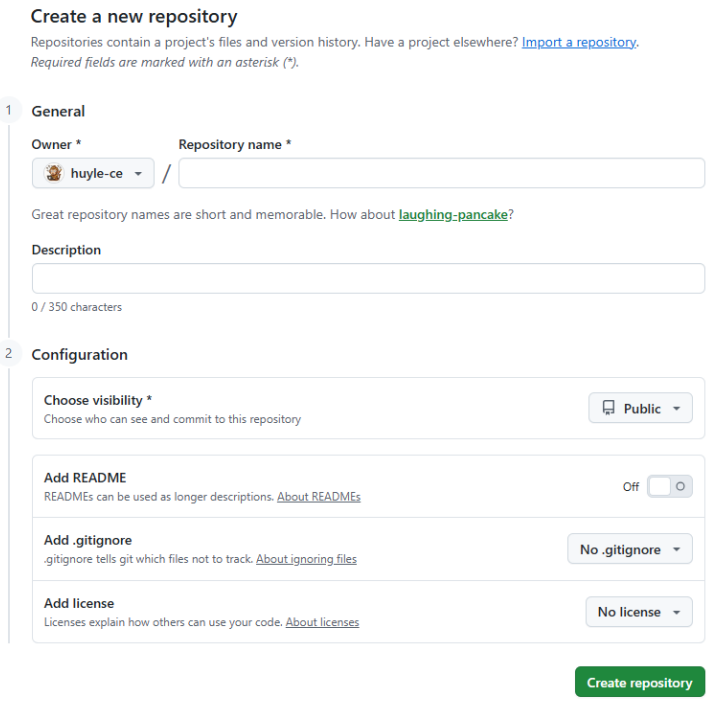
The very first thing to do is creating a repository ('repo' for short) where your project is managed. To do this, open your user navigation menu (the round icon on the top-right corner), then click `Repositories`, see the picture below:



And then, click the green `New` button:



Finally, configure your repo by filling the boxes, with the important thing being "Choose visibility" of "Public/Private". After that, click "Create repository" to create the repo

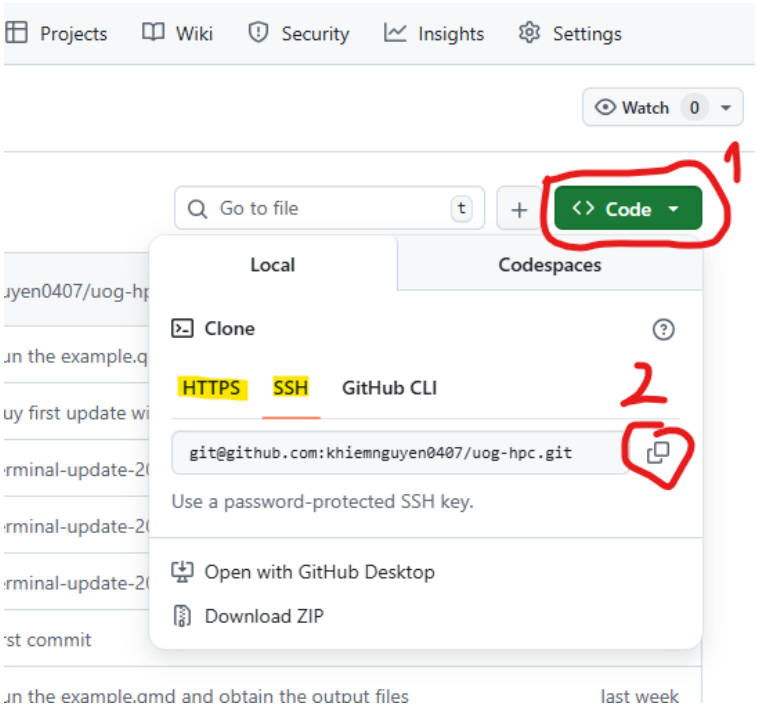


9.3 Connect to a GitHub's repository

Basically, there are two cases to get connected to a GitHub's repo, including connecting to your own GitHub repo and connect to a repo that you are invited by your collaborator. Both cases are done in a very similar way, through either `HTTPS` or `SSH` protocol.

`HTTPS` and `SSH` are protocols used by Git to communicate with remote repositories (e.g., GitHub), with the difference mainly lying in authentication, security mechanism, and workflow convenience.

Either you are connecting to your own repo or to the repo you were invited, the first thing to do is go to that repo to get the directory URL. Depending on which method you prefer, you will select `HTTPS` or `SSH` to get the link. At this point, we work through `HTTPS` and `SSH`, while leave `GitHub CLI` in later topic.



9.3.1 HTTPS - HyperText Transfer Protocol Secure (good for quick or temporary access)

By doing this way, when GitHub needs authentication, you will have to use "**Personal Access Token (PAT)**" (*this setup is left for you to find how to do yourself*).

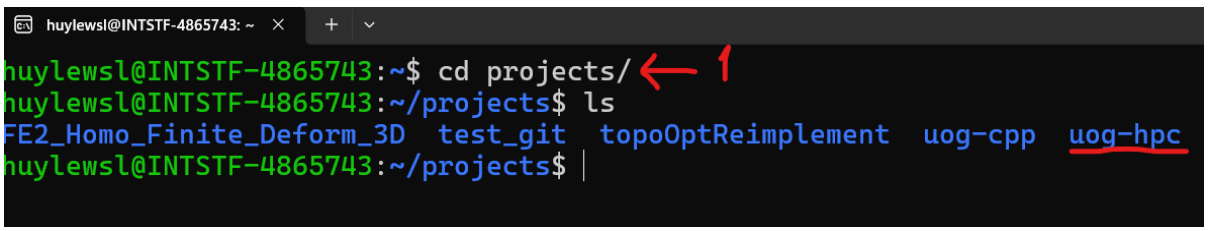
Using the copied `HTTPS` URL, you can now `clone` the repo to your local directory where your want to put your project in your local machine. In you terminal, type

```
git clone <your copied HTTPS URL>
```

For example,

```
git clone https://github.com/khiemnguyen0407/uog-hpc.git
```

To test the job, in your terminal, at the folder you keep the cloned repository, type `ls` to see whether the repo was sucessfully cloned to you local repo. See the following example of my case, first I go to my `projects` folder where I put the cloned repo of `uog-hpc`, and then `ls`.



9.3.2 SSH - Secure Shell (good practice and recommended)

We will go in detail of how to get connected through `SSH` protocol. It is considered as a good practice and is recommended for a long-term use of GitHub. Some of the advantages are:

- No repeated login
- Very convenient for frequent Git use
- More natural workflow for developers
- Preferred for automated systems (HPC, CI, etc.)

SSH authenticates by using the so-called "**public–private key cryptography**", which generates a "`private key`", only kept on your machine, and a "`public key`", kept on both your machine and on you GitHub account. Followings are steps to setup the SSH authentication.

This tutorial will adhere to SSH connection.

9.3.2.1 Step-by-step setup for SSH authentication:

Step 1: Generate the key: in your terminal type, and fill it with your email which you used to register your GitHub account

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

For example, my case is

```
ssh-keygen -t ed25519 -C "huy.lequang.ce@gmail.com"
```

Then, press Enter to choose the default location for storing the key (you can choose your own custom location, but we should do so unless you do have a purpose on this)

Next, enter your `passphrase` (it is like a password) to protect your `private key`. It is optional, but it provides one more layer to protect your `private key` from being exposed to someone else. (Imagine some one use your machine and open the file to have your `private key`)

Step 2: Start the SSH agent: in your terminal type

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

Step 3: Copy the `public key` in order to add to your GitHub account. In your terminal, type

```
cat ~/.ssh/id_ed25519.pub
```

, and then copy the whole line.

Step 4: Add the key to your GitHub account: go to your GitHub settings (click on your account icon) –> go to "SSH and GPG keys" –> click "New SSH key" to add the key by pasting the copied content at Step 3. Note that, GitHub allows **multiple SSH keys per account**, so if you have other authentications (other machine), you can leave them untouched, and add the new SSH key.

Step 5: Test the authentication: in your terminal, type

```
ssh -T git@github.com
```

, then you should see something like:

```
Hi huyle-ce! You've successfully authenticated, but GitHub does not provide shell access.
```

Step 6: now, you can clone a repository that you have permission to clone, which can be either an invited or your own one. First, go to GitHub and copy the SSH address from the repository you want to clone, then clone it by command line:

```
git clone git@github.com:path/repository.git
```

Now, you can work with your local and GitHub repositories by pulling, pushing, and so on.

9.3.2.2 Understanding about public–private key cryptography:

After going through the procedure for setting up the SSH authentication above, it is worth to have a sense on how it works.

When doing `git clone git@github.com:user/repo.git`, the SSH protocol runs to establish a SSH channel to the GitHub's directory. Next, the SSH sends the `public key` to GitHub, and then GitHub compares this particular key with the `stored public keys`.

So, GitHub knows that this connection (or loosely speaking, this machine) is trying to authenticates as *account X* which holds the matched `public key`. Through the SSH channel, the `public key` of the account X is checked with the `private key` of the machine. If they match, the account is authenticated to be authorized by the machine, and so get connected.

When you try to clone a repository via SSH protocol, GitHub will check whether your account has permission for cloning this repository.

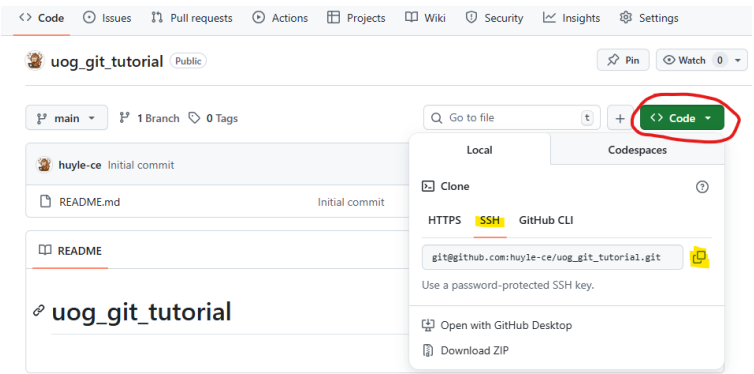
There are actually many complicated technical features behind the scene, but for a benign understanding, it can be thought of `private key` as a "key" and the `public key` as a "lock". Your machine knows both the key and the lock, while your GitHub's account holds the lock. By connecting to your GitHub account through SSH, you are asking GitHub to find the account associated with this "lock" and then use the "key" to open that "lock". If someone else can have your `public key` in their hand, and try to ask GitHub to connect themselves to your account using that `public key`, then they cannot as their machine does have the "key" - `private key` - to open the "lock".

10 Get started

This section aims at getting you acquainted with Git through a simple workflow where you will work with your own GitHub's repo. Now, let's put into practise what we have gone throught above.

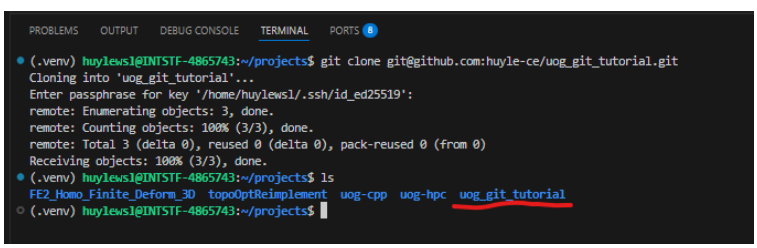
Start from `cloning` a remote repository

First, we need to clone our repo which we already created from GitHub (provide your "passphrase" if it asks you to do so):

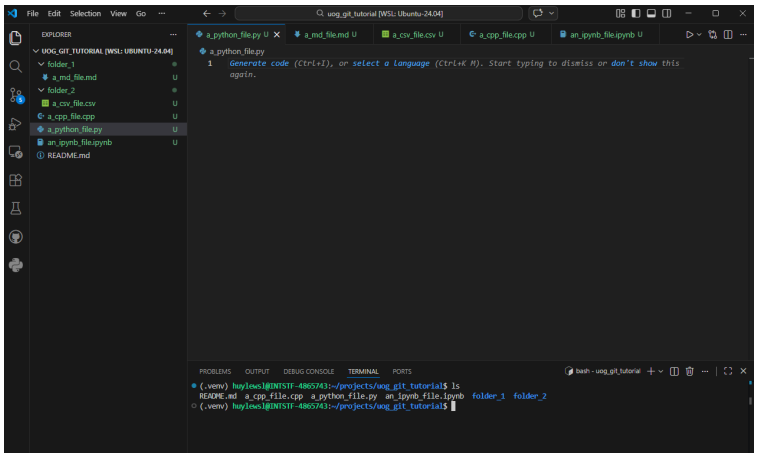



```
git clone git@github.com:huyle-ce/uog_git_tutorial.git
```

Go to your terminal , type `ls` to test whether it has done fine!



Now, we can `cd` to the repo and then type `code .` to open VS Code in this repo. It will lead you to a fresh and blank VS Code window. Hereon, you can perform tasks in exactly the same as you normally do with VS Code. Let's create some folders and files so that we can have things to do examples.



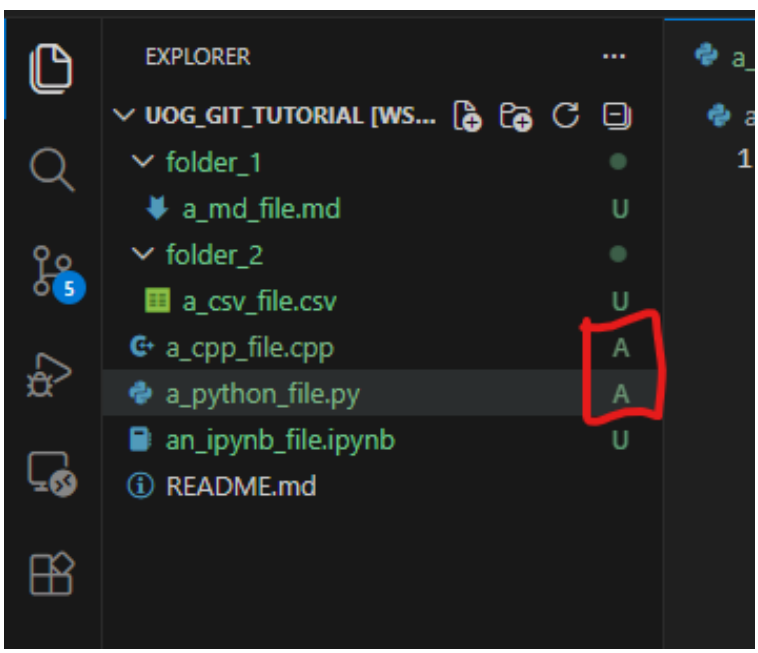
You can see the newly created folders and files appear to be green and with letter "U" on the right. This means those files are on `Working Directory` and are not tracked by `Git`, so every change and modification are basically saved on local system. Now, let us do the basic workflow step-by-step.

1. Add the files to `Staging Area`. You can add specific files or all the files.

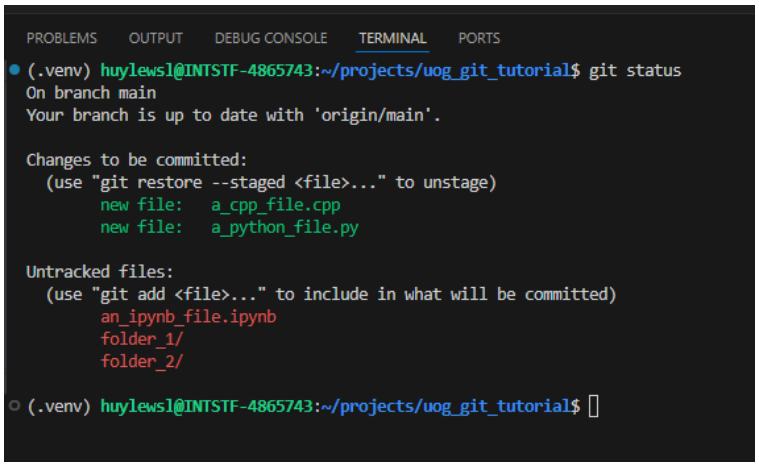
```
# Add specific files
git add a_cpp_file.cpp a_python_file.py

# Or, add all the files
git add .
```

The statuses of the added files are now changed into "A"



A handy command of a git user is `git status` which is to show the status of the current file. The following picture show the status of the files in which `a_cpp_file.cpp a_python_file.py` were added while the others were not.

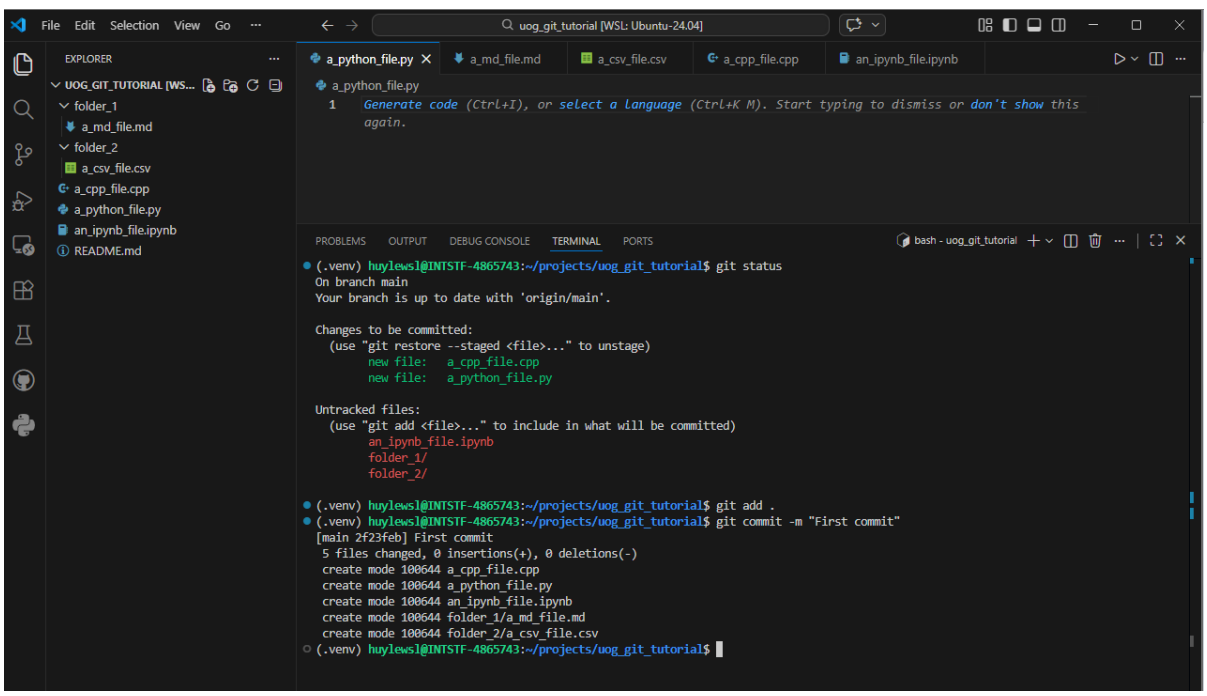


Now, we want to do it for all the files.

2. Next, we need to `commit` the added files to the so-called `Local Repository`. By doing this, Git takes everything currently in the staging area and permanently records it as a new version in the history. You can think of the `commit` action as the `save` action that you normally do when using your computer.

```
git commit -m "First commit"
```

Then you can see all the files appearing in your explorer become white and some information is provided in your terminal.



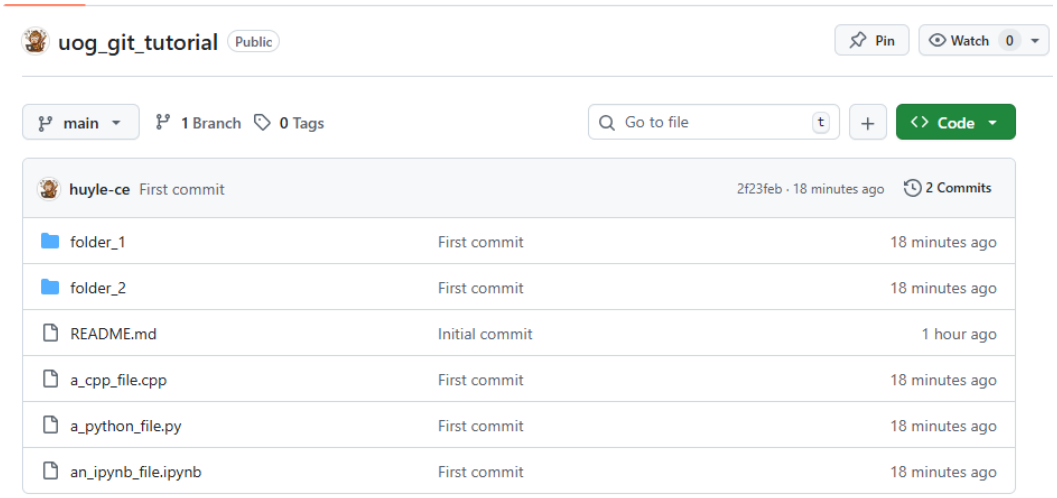
3. Up to this point, your changes are on your `main branch` and are just recorded by Git on your `Local Repository`, nothing has touched your GitHub repo. What we need to to next is `push` it to your remote repo which is essentially your GitHub's repo.

```
git push
```

What Git actually does is that it push the `main` branch to the `origin` of your remote repo. Recall that the first step is cloning the remote repo to your local machine. Through this process, a "channel" between the `main` and `origin` is established. We will go through branches in Git in detail later.

```
(.venv) huylewsl@INTSTF-4865743:~/projects/uog_git_tutorial$ git branch
* main
(.venv) huylewsl@INTSTF-4865743:~/projects/uog_git_tutorial$ git push
Enter passphrase for key '/home/huylewsl/.ssh/id_ed25519':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 22 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 474 bytes | 474.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:huyle-ce/uog_git_tutorial.git
   685e995..2f23feb  main -> main
(.venv) huylewsl@INTSTF-4865743:~/projects/uog_git_tutorial$
```

Now, you can revisit your GitHub repo on your web browser, refesh the page and you will see all the files were streamed to there.



Start from local repository